

CMSRN AT A GLANCE

Owner	MSNCB (Medical-Surgical Nursing Certification Board)
Items	150 multiple choice — 125 scored, 25 unscored
Time	3 hours
Pass mark	Standard score of 95, roughly 71% of the 125 scored items
Standard setting	Modified Angoff, set by subject matter experts
Result	Preliminary pass/fail on screen at the end of the exam
Blueprint	2023 CMSRN blueprint, in force since 15 May 2023

BLUEPRINT DOMAINS

Domain	Weight	Items
Patient/Care Management	32%	40
Nursing Teamwork and Collaboration	21%	26
Elements of Interprofessional Care	17%	21
Holistic Patient Care	15%	19
Professional Concepts	15%	19

TIP Roughly a third of the exam is patient and care management, but teamwork, interprofessional care and professional concepts together are more than half. Study judgement and communication as hard as pathophysiology.

LABS TO KNOW — RANGES VARY BY LAB

Test	Usual adult range
Sodium	135–145 mEq/L
Potassium	3.5–5.0 mEq/L
Calcium (total)	8.5–10.5 mg/dL
Magnesium	1.5–2.5 mg/dL
BUN / creatinine	7–20 / 0.6–1.2 mg/dL
Haemoglobin (F / M)	12–16 / 14–18 g/dL
Platelets	150–400 ×10 ⁹ /L
White cell count	4.5–11 ×10 ⁹ /L
INR (no anticoagulant)	0.8–1.2
Fasting glucose / HbA1c	70–99 mg/dL / under 5.7%

ELECTROLYTE PATTERNS

- Low potassium: flat T waves, U waves, weakness; hypokalaemia raises the risk of digoxin toxicity.
- High potassium: peaked T waves can progress to a widened QRS; with ECG changes, stabilise the myocardium with IV calcium, then shift and remove potassium per protocol.
- Low sodium: headache, confusion and seizures; severe symptomatic hyponatraemia needs prompt controlled hypertonic saline while still avoiding overly rapid overall correction.
- Low calcium: tingling, tetany, positive Chvostek and Trousseau signs.
- Low magnesium: tremor, torsades risk; it also keeps potassium low.
- Check kidney function before blaming intake for any electrolyte result.

CALCULATIONS YOU MAY BE ASKED TO RUN

Mean arterial pressure	$MAP = (SBP + 2 \times DBP) \div 3$
Infusion rate	$mL/hr = \text{total volume} \div \text{hours}$
Gravity drip rate	$ggt/min = (\text{volume} \times \text{drop factor}) \div \text{minutes}$
Dose by weight	Ordered dose per kg × weight in kg Convert pounds to kg by dividing by 2.2
Tablets or mL to give	$(\text{Desired} \div \text{on hand}) \times \text{quantity}$
Body mass index	$\text{Weight (kg)} \div \text{height (m)}^2$
Fluid balance	Total intake – total output Include drains and emesis

FINDINGS THAT NEED ESCALATION

- A new or worsening change in level of consciousness.
- Systolic pressure under 90 mmHg, or a sustained fall from the patient's baseline.
- Respiratory rate under 8 or over 28, or rising sedation on opioids.
- Urine output under about 0.5 mL/kg/hr over several hours.
- New chest pain, new shortness of breath, or a new unilateral swollen leg.
- Any patient the nurse is simply worried about — that counts.



WOUNDS, DRAINS AND DEVICES

- Describe a wound by size, depth, tissue type, exudate and surrounding skin.
- Serous is thin and clear; sanguineous is bloody; purulent suggests infection.
- Dehiscence: support the wound and notify the surgical team; if evisceration occurs, cover exposed organs with sterile saline-moistened dressings and get urgent help.
- A new ostomy should be pink and moist; dusky or dry needs urgent review.

HIGH-YIELD MED-SURG RECALL

- Heart failure: daily weight is a sensitive trend for fluid retention; read it with symptoms, examination and intake/output.
- COPD: watch for rising CO₂ and a falling level of consciousness, not just oxygen saturation.
- Sepsis: take cultures and a lactate promptly, ideally before antibiotics but without delaying them; start antimicrobials promptly and begin fluids immediately for hypoperfusion.
- Stroke: time of last known well drives every treatment decision.
- DKA: start fluids and check potassium before insulin; if potassium is below about 3.5 mmol/L, replace it and delay insulin, otherwise start insulin and add potassium as levels fall.
- Post-op: pulmonary hygiene and early progressive mobility reduce atelectasis, clots and deconditioning; tailor them to risk and orders.
- Opioids: sedation rises before the respiratory rate falls — score sedation.

SAFETY AND PREVENTION

- Two identifiers before every medication, procedure and specimen.
- Read back and verify any verbal or telephone order.
- Reassess fall risk on admission, after a change in status, and after any fall.
- Reposition, offload heels and keep skin dry to prevent pressure injury.
- Hand hygiene before and after every patient contact, gloves or not.

TEAMWORK AND INTERPROFESSIONAL CARE

- Hand-off with a structured tool and an opportunity to ask questions.
- Delegate only tasks permitted by the delegatee's scope, competence, patient condition, employer policy and nurse practice act; the RN retains nursing judgment, overall assessment, planning, supervision and evaluation, and unstable patients require RN-level assessment and oversight.
- The five rights of delegation: task, circumstance, person, direction, supervision.
- Escalate with the chain of command when a patient is unsafe.
- Discharge planning starts at admission and is interdisciplinary — nursing, case management, social work and the wider team.
- Teach back confirms the patient actually understood the education.

PROFESSIONAL CONCEPTS

- Scope of practice and your state nurse practice act come before unit habit.
- Evidence-based practice: ask, search, appraise, apply, evaluate.
- Advance directives and informed consent belong to the patient.
- Protect confidentiality in person, on paper and in the record.
- Report incidents factually; the record is not the place for blame.
- Advocate when the plan of care and the patient's stated goals do not match.

HOLISTIC CARE: PAIN, MOBILITY, NUTRITION

- The patient's self-report is the standard for pain intensity.
- Use a behavioural scale only when the patient cannot self-report.
- Multimodal analgesia and non-drug measures reduce opioid need.
- Screen nutrition on admission; involve dietetics for poor intake or wounds.
- Address spiritual, cultural and family needs as part of the plan, not as an extra.

TEST-DAY PLAN

1. About 72 seconds per item — flag and move rather than stall.
2. Assess before you intervene, unless the stem describes an emergency.
3. Airway, breathing, circulation, then safety, then comfort.
4. Pick the least invasive nursing action within your scope.
5. Answer every item; there is no penalty for a guess.

TIP Study aid only — not clinical guidance. Reference ranges vary by laboratory; confirm the current blueprint and scoring in the MSNCB CMSRN Certification Handbook.

